Parameterised model run report v1.0.0

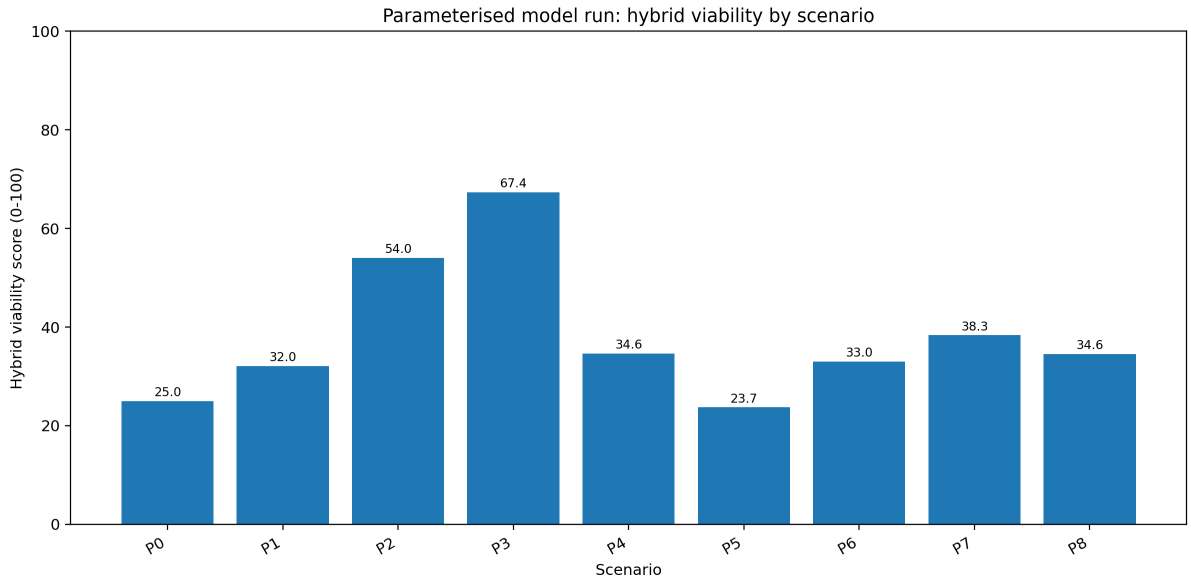
Source-informed demonstrative parameterisation; not an empirically calibrated predictive model.

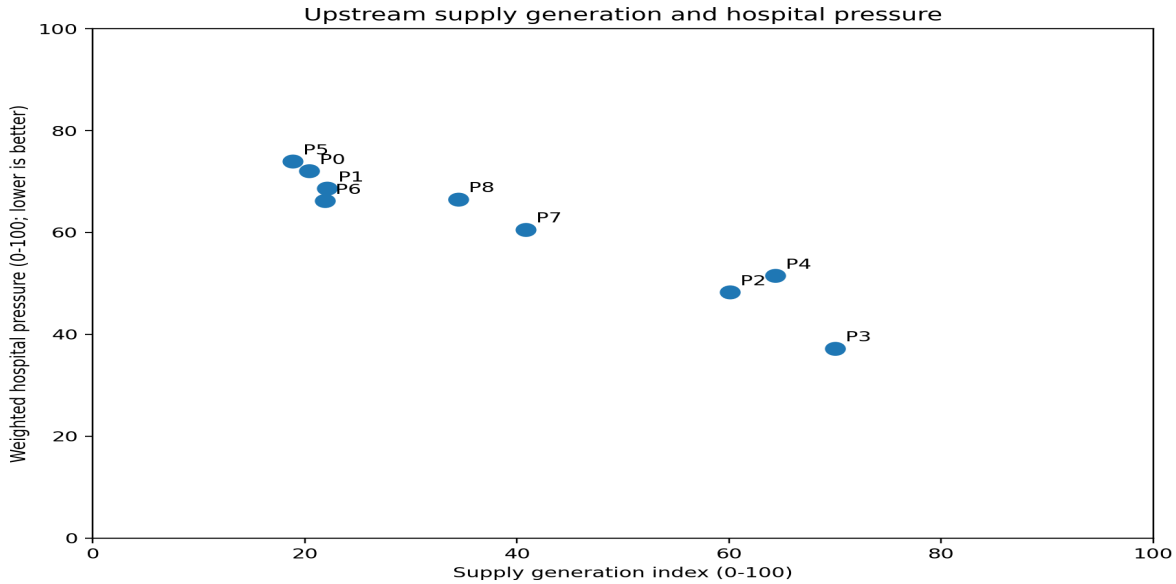
Executive summary

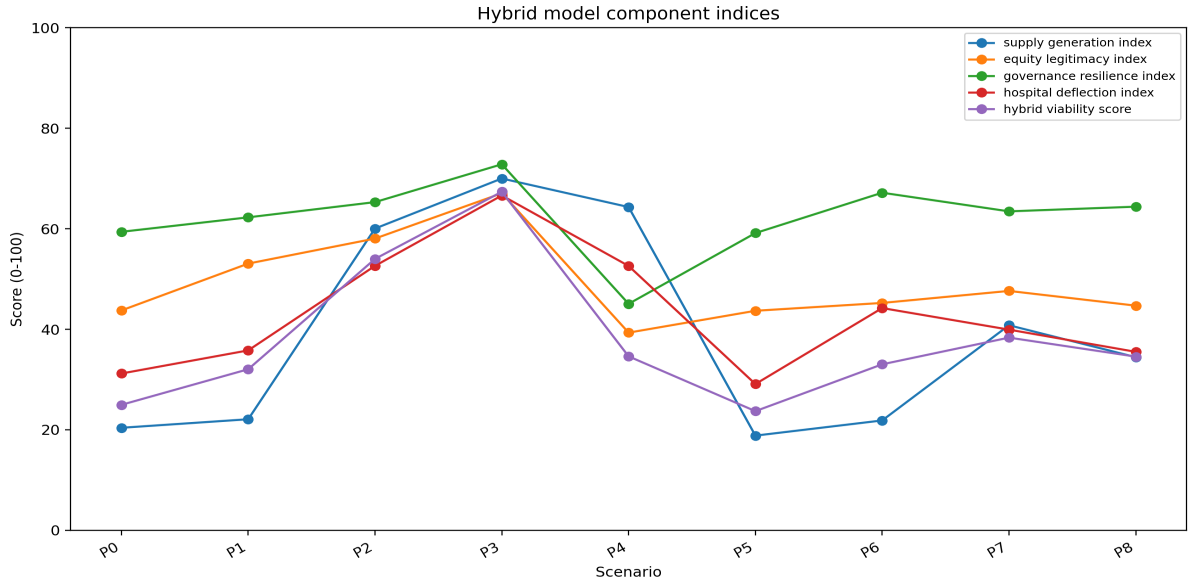
This version identifies parameters, attaches public-source inputs, updates and reruns the game, hybrid, uncertainty and MCDA layers, then packages the results as plots and tables. The strongest conclusion remains: demand-driven within rules; not demand-driven without rules.

Hybrid results

scenario id	scenario name	hybrid viability score	supply generation index	hospital deflection index	weighted hospital pressure	fiscal gaming risk index	interaction penalty
P0	Current source-informed baseline	24.97	20.40	31.20	72.04	14.27	14.06
P1	Capitation reweighting plus access target	32.05	22.10	35.80	68.64	12.90	11.81
P2	National Primary Care Benefits Schedule	54.00	60.04	52.62	48.24	16.60	5.88
P3	Full upstream access architecture	67.36	70.00	66.62	37.22	10.84	3.09
P4	Benefits without adequate controls	34.61	64.32	52.60	51.50	41.82	15.15
P5	ACC activity constraint shock	23.71	18.84	29.10	73.96	14.13	14.16
P6	Ambulance alternatives only	33.03	21.86	44.22	66.20	12.46	12.79
P7	Scope-enabled provider supply only	38.34	40.80	39.93	60.58	16.45	10.83
P8	Direct claims and PHO intermediation reform only	34.56	34.45	35.51	66.53	16.34	11.21

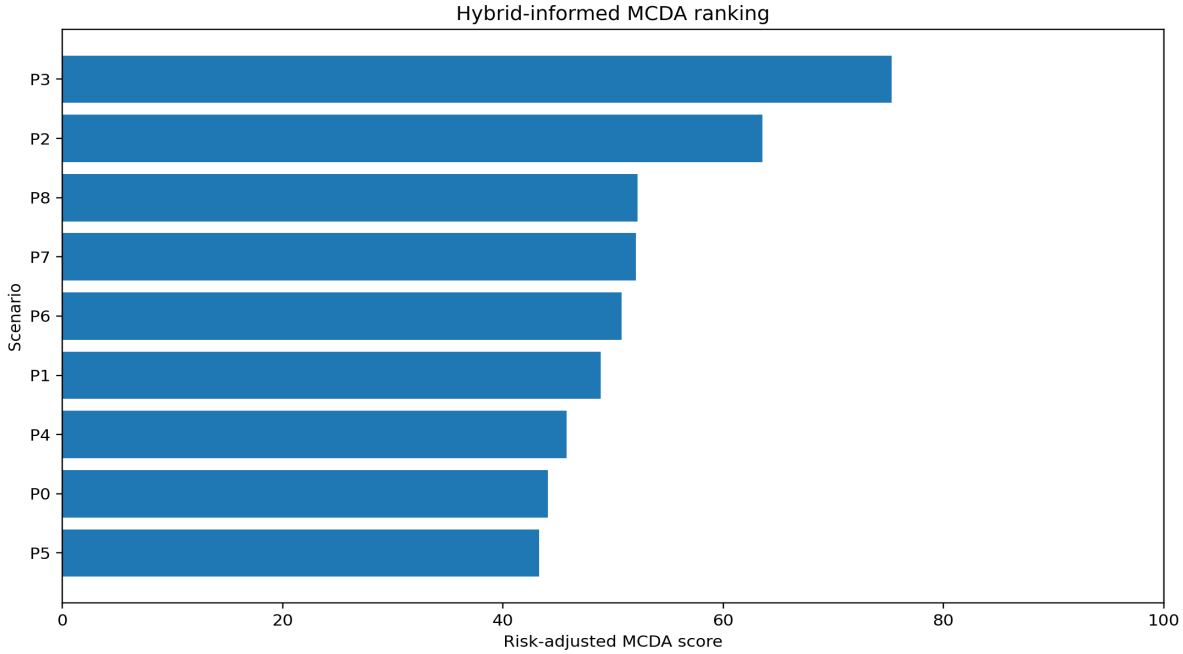

Parameterised model run: hybrid viability by scenario
Hybrid viability score (0-100)
Scenario
0
20
40
60
80
100
25.0
32.0
54.0
67.4
34.6
23.7
33.0
38.3
34.6
P0
P1
P2
P3
P4
P5
P6
P7
P8


Upstream supply generation and hospital pressure
Weighted hospital pressure (0-100; lower is better)
Supply generation index (0-100)
0
20
40
60
80
100
P5
P0
P1
P6
P8
P7
P4
P2
P3


Hybrid model component indices
supply generation index
equity legitimacy index
governance resilience index
hospital deflection index
hybrid viability score
Score (0-100)
0
20
40
60
80
100
P0
P1
P2
P3
P4
P5
P6
P7
P8
Scenario

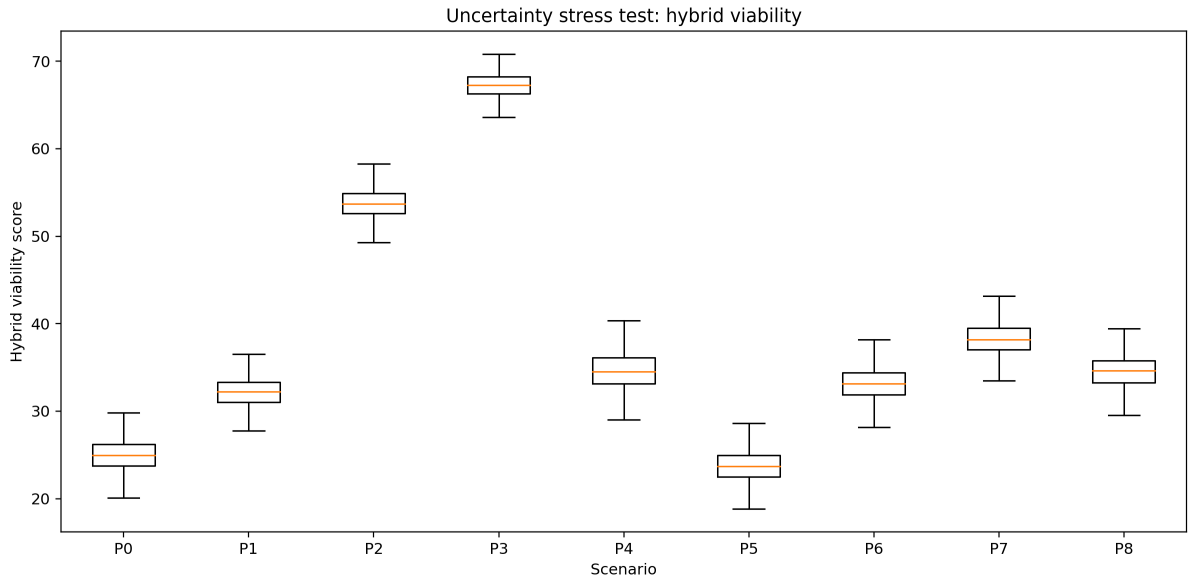
Hybrid-informed MCDA

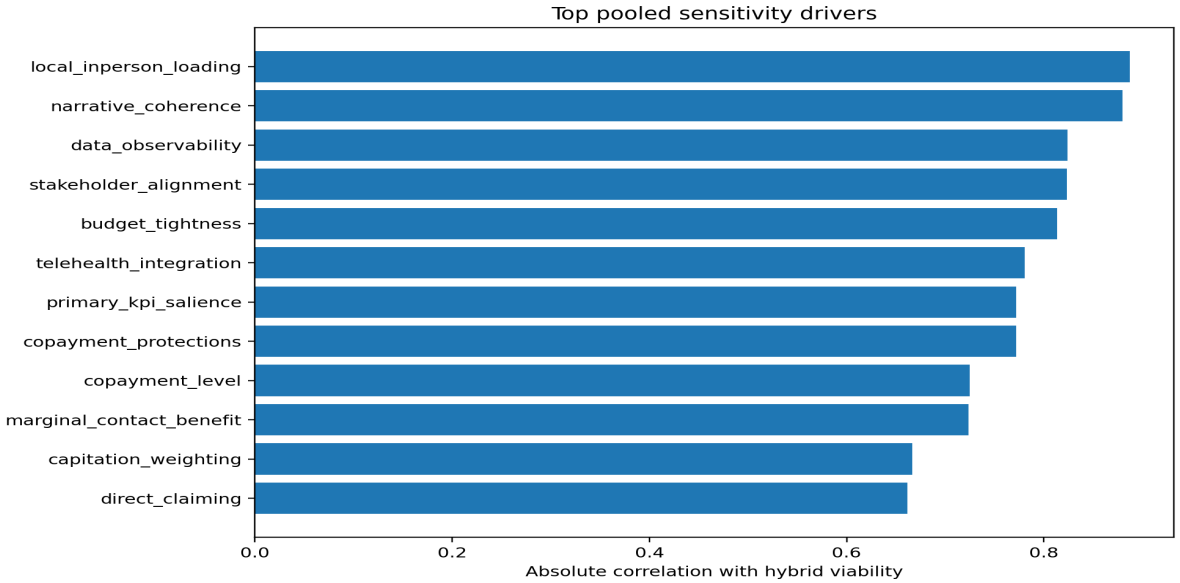
rank	scenario id	scenario name	weighted score before penalty	risk penalty	risk adjusted mcda score
1	P3	Full upstream access architecture	77.34	2.04	75.30
2	P2	National Primary Care Benefits Schedule	66.79	3.24	63.54
3	P8	Direct claims and PHO intermediation reform only	55.96	3.74	52.22
4	P7	Scope-enabled provider supply only	55.76	3.71	52.05
5	P6	Ambulance alternatives only	54.06	3.27	50.78
6	P1	Capitation reweighting plus access target	52.11	3.25	48.86
7	P4	Benefits without adequate controls	53.96	8.21	45.75
8	P0	Current source-informed baseline	47.75	3.69	44.06
9	P5	ACC activity constraint shock	46.96	3.68	43.28

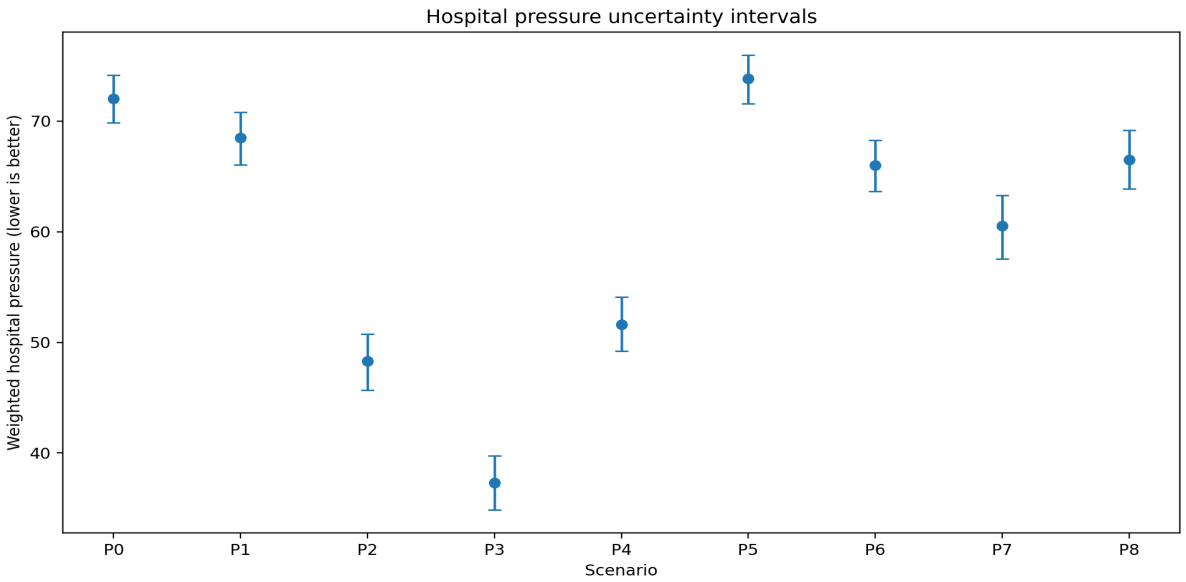


Uncertainty and sensitivity

scenario id	scenario name	hybrid viability score mean	hybrid viability score p05	hybrid viability score p95	weighted hospital pressure mean	weighted hospital pressure p05	weighted hospital pressure p95
P0	Current source-informed baseline	24.96	21.92	27.91	72.03	69.86	74.17
P1	Capitation reweighting plus access target	32.17	29.24	34.90	68.48	66.05	70.82
P2	National Primary Care Benefits Schedule	53.75	51.02	56.47	48.31	45.67	50.74
P3	Full upstream access architecture	67.26	64.86	69.57	37.29	34.84	39.74
P4	Benefits without adequate controls	34.57	31.27	37.87	51.61	49.19	54.08
P5	ACC activity constraint shock	23.74	20.70	26.91	73.86	71.56	75.99
P6	Ambulance alternatives only	33.17	30.34	36.21	66.03	63.67	68.27
P7	Scope-enabled provider supply only	38.26	35.16	41.57	60.51	57.55	63.30
P8	Direct claims and PHO intermediation reform only	34.50	31.18	37.58	66.50	63.88	69.19

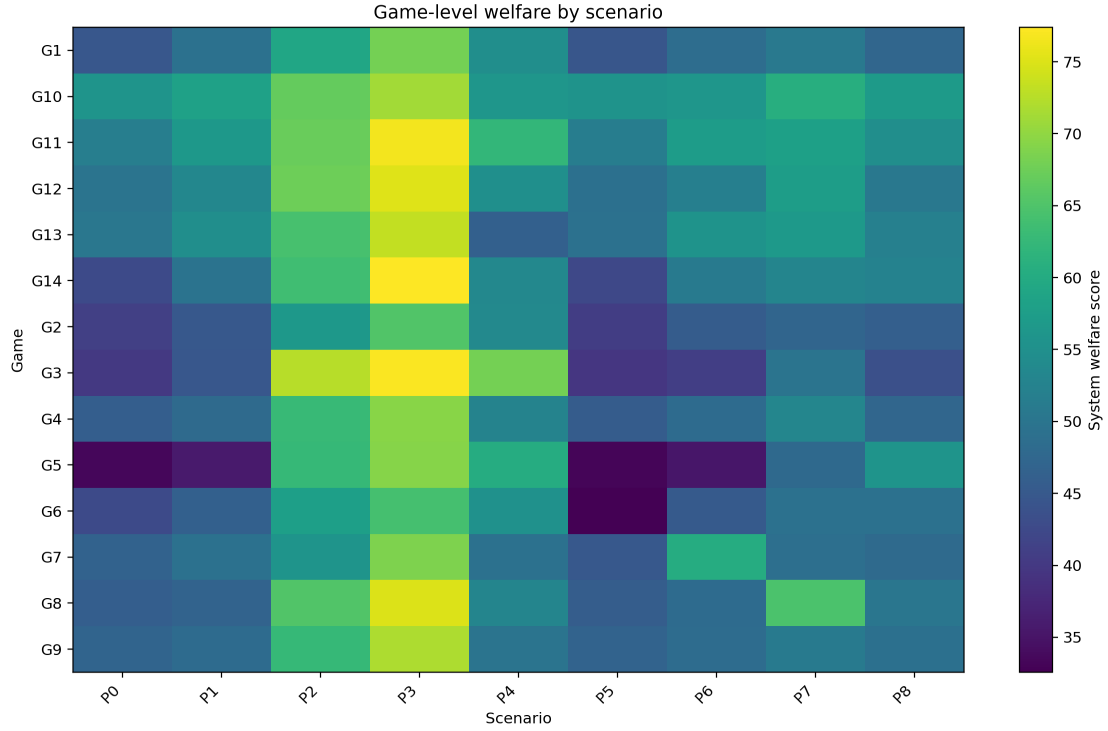



Top pooled sensitivity drivers
local_inperson_loading
narrative_coherence
data_observability
stakeholder_alignment
budget_tightness
telehealth_integration
primary_kpi_salience
copayment_protections
copayment_level
marginal_contact_benefit
capitation_weighting
direct_claiming
0.0
0.2
0.4
0.6
0.8
Absolute correlation with hybrid viability


Hospital pressure uncertainty intervals
Weighted hospital pressure (lower is better)
70
60
50
40
P0
P1
P2
P3
P4
P5
P6
P7
P8
Scenario

Game-level results

scenario id	scenario name	access score	hospital pressure	gaming risk	system welfare
P0	Current source-informed baseline	38.10	72.07	20.42	45.58
P1	Capitation reweighting plus access target	41.41	68.66	19.60	48.97
P2	National Primary Care Benefits Schedule	66.69	48.81	18.93	63.17
P3	Full upstream access architecture	76.44	37.63	15.38	71.61
P4	Benefits without adequate controls	66.62	52.52	34.04	54.96
P5	ACC activity constraint shock	36.13	74.05	20.37	44.41
P6	Ambulance alternatives only	42.39	66.05	18.01	49.50
P7	Scope-enabled provider supply only	51.74	60.87	20.78	53.46
P8	Direct claims and PHO intermediation reform only	45.88	66.51	19.25	50.25



Parameter inputs

parameter	domain	source value	current normalised value	evidence level	confidence 0 to 1
marginal_contact_benefit	funding	Capitation is core general practice funding; ACC pays activity/contract/regulation-based injury treatment.	0.10	public architecture + modelling judgement	0.65
capitation_weighting	funding	Formula introduced in 2002; new approach expected from 1 July 2026.	0.35	public fact + modelling judgement	0.80
scope_flexibility	workforce/scope	ACC contributes to consultation rates and procedures for GPs, nurses and nurse practitioners.	0.35	public architecture + modelling judgement	0.55
pho_transaction_cost	administration	Primary care data governance and SLM processes involve PHOs; closed-books availability is reported as a primary care indicator.	0.60	public architecture + hypothesis	0.45
primary_kpi_salience	accountability	>80% of people can access an appointment with a general practice provider within one week.	0.40	public fact + modelling judgement	0.80
ambulance_kpi_salience	accountability	Monthly reports show response time KPIs, demand, quality of service and call type.	0.45	public fact + modelling judgement	0.75
hospital_political_penalty	political economy	Health NZ plans and commissions hospital, primary and community services.	0.90	strategic modelling hypothesis	0.45
data_observability	data	Participating practices share encounter and appointment data from April 2026; first focus is the seven-day access target.	0.45	public fact + modelling judgement	0.85
copayment_level	consumer price	Adults cost barrier 14.9%; Pacific adults 25.1%; M■ori adults 18.7%.	0.60	public empirical input	0.85
copayment_protections	consumer price	Cost barrier persisted at 14.9% nationally; Pacific 25.1%, M■ori 18.7%; disabled adults had more access barriers.	0.45	public empirical input + modelling judgement	0.70
equity_program_strength	equity	Pacific adults had highest GP cost barrier; disabled adults had more barriers including owing money, transport and support person.	0.60	public empirical input + modelling judgement	0.60
telehealth_scale	digital access	Online GP Care is mentioned as a performance-reporting example; exact volumes not yet in public calibration pack.	0.65	public architecture + modelling judgement	0.40
telehealth_integration	digital access	Shared Digital Health Record and NPCD are being developed; participation and data processes are evolving.	0.45	public architecture + modelling judgement	0.55
local_inperson_loading	rural/local supply	25.5% of adults reported appointment wait time as a barrier.	0.25	public empirical input + modelling judgement	0.60
ambulance_alternative_funding	ambulance	Emergency ambulance services are commissioned on behalf of Health NZ and ACC; medical and injury funding split.	0.35	public architecture + modelling judgement	0.65
acc_activity_funding	ACC	ACC contributes to consultation rates and specific treatment/procedure costs for GPs, nurses and nurse practitioners.	0.70	public architecture + hypothesis	0.55
acc_constraint	ACC	How ACC pays depends on service type and whether provider works under regulations, contract or purchase order.	0.10	public architecture + modelling hypothesis	0.45
budget_tightness	fiscal	Health NZ is responsible for planning and commissioning hospital, primary and community health services.	0.85	strategic modelling hypothesis	0.50

parameter	domain	source value	current normalised value	evidence level	confidence 0 to 1
safety_governance	governance	ACC payment conditions include appropriate qualifications, current practising certificate, quality and documentation.	0.75	public architecture + modelling judgement	0.70
gaming_controls	governance	Advisory/governance groups will advise target definition, implementation and balancing measures.	0.60	public architecture + modelling judgement	0.65
direct_claiming	payment platform	Capitation is not paid directly to patients; PHO/governance structures remain central to data/funding arrangements.	0.05	public architecture + policy option	0.60
stakeholder_alignment	political economy	Not yet measured.	0.35	unvalidated judgement	0.25
narrative_coherence	political economy	Not yet measured.	0.30	unvalidated judgement	0.25

Limitations

The run is source-informed and auditable, but not empirically calibrated. Results should be treated as prioritisation and decision-support outputs until calibrated against administrative and stakeholder data.